

TeleTouch : Human-to-Human Teleoperation with Bare Finger Tactile Sensing and High-Resolution Electro-tactile Display

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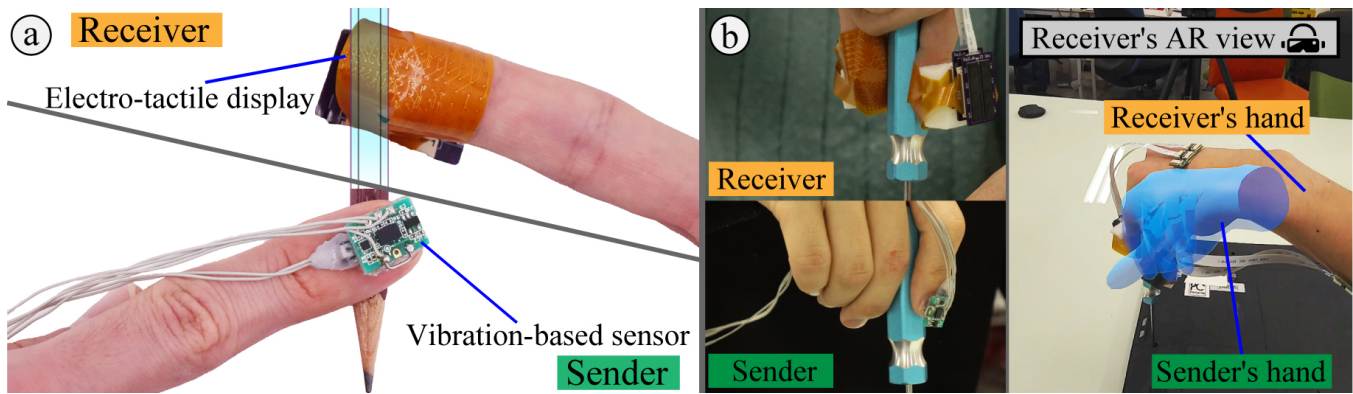


Figure 1: (a) Sensors capture edge direction, force, and contact point (bottom), and an electro-tactile display reproduces the sensation based on the results of human tactile acuity measurements (top); (b) Example applications.

Abstract

This paper introduces TeleTouch, a system that enables human-to-human tactile transmission between remote locations. By capitalizing on the innate dexterity and sensitivity of human touch, TeleTouch facilitates natural and intuitive remote interactions. The system comprises two primary components: (1) a vibration-based tactile sensor that operates without covering fingertip, and (2) a high-resolution electro-tactile display that replicates the spatial resolution of human fingertip perception. Additionally, the integration of an AR head-mounted display (AR-HMD) enables real-time visual sharing of hand movements and positions, thereby enhancing situational awareness and facilitating coordinated tactile interaction.

* Authors contributed equally to this research.

CCS Concepts

• **Human-centered computing** → **Haptic devices**; *Virtual reality*; *Mixed / augmented reality*.

Keywords

teleoperation, haptics, tactile sensing, electro-tactile, fingertips

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1 Introduction

This project aims to realize teleoperation by transmitting tactile information between remote human operators (Figure d1). In many teleoperation scenarios, a robot is deployed in a real-world work environment where environmental data—including tactile information—is captured and relayed as sensory feedback to the remote operator [Trinitatova and Tsetserukou 2023]. On the other hand, despite current advances in robotics, the dexterity of human hands

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remains superior to that of robots. Therefore, the most robust teleoperation may be achieved if the tactile and motor capabilities of a remote operator can be effectively conveyed to a local operator.

Implementing direct human-to-human tactile transmission between remote locations requires the development of two key technological elements. First, it is necessary to develop a tactile sensing method for real work environments that does not interfere with the user's (hereinafter referred to as "Sender") natural sense of touch. As mentioned above, dexterous human manipulation relies on information regarding the direction in which a tool is grasped. Therefore, it is essential to capture not only the overall force applied to the fingertip but also the spatial distribution of skin contact. Since the remote user is a human rather than a robot, integrating a highly accurate tactile distribution sensor into the fingertip is challenging [Johnson et al. 2011]. Moreover, wearing a glove-type tactile sensor may compromise the Sender's natural sense of touch.

The second key element is the development of a tactile display system that presents tactile information with sufficient spatial resolution to the remote user (hereinafter referred to as the "Receiver"). Sobue et al. [Sobue et al. 2025] reported that tactile acuity of the index finger increases gradually from the finger pad toward the tip. Tactile displays that match this natural gradient of tactile acuity are therefore necessary.

In this project, TeleTouch integrates a vibration-based tactile sensor that leaves the fingertip uncovered with a high-resolution electro-tactile display replicating human spatial acuity. An AR head-mounted display then streams real-time hand motion and position, enabling shared visual context and coordinated tactile interaction.

2 Technical Elements

2.1 Vibration-based Sensing

The sensing module is depicted in Figure 2 (a). It comprises a vibration motor (LBV10B-009, Nidec) and a set of acceleration and angular velocity sensors (BMX055, BOSCH) to estimate finger contact information [Kon et al. 2025]. Using an eccentric motor, the finger is induced to undergo two-dimensional (2-DOF) circular vibrations. When the vibrating finger contacts an object, the vibration characteristics—such as amplitude and phase—vary according to the object's properties. The vibration data are represented as scatter plots of acceleration and angular velocity for the X, Y, and Z axes. These plotted points (x_n, y_n) are transformed into polar coordinates (r, θ), generating vectors (Figure 2 (b)). The resulting feature vector is utilized to estimate key object properties through classification by a Support Vector Machine (SVM).

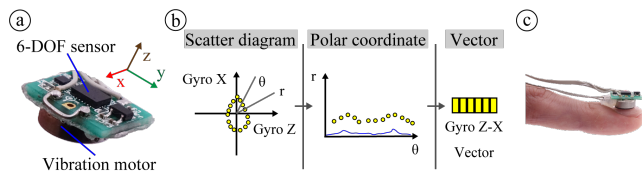


Figure 2: (a) Appearance of the device; (b) Data processing for estimation (here is an example of generating a Gyro Z-X vector); (c) Example of an attached view.

A significant advantage of this method is that it acquires tactile information without obstructing the finger pad's direct contact with the object. By employing active sensing through vibration, the system simultaneously captures crucial parameters—including the direction of the object's edge, the contact position, and the pressing force—necessary for dexterous fingertip manipulation, and relays this tactile data to a remote user.

2.2 Electro-tactile Display

The tactile presentation unit of the tactile transmission device mainly consists of an electro-tactile display (Figure 3 (a)), which was developed based on the results of tactile acuity measurements [Sobue et al. 2025]. In this design, the electrode distance gradually varied along the fingertip: 2.5 mm at the proximal (joint) side and 1.25 mm at the tip of the index fingertip. Furthermore, the inter-electrode spacing on the lateral sides of the finger pad is considerably wider than that in the central region of the finger pad (Figure 3 (b)). All electrodes have a diameter of 1 mm, and they are mounted on a flexible circuit board that is divided into front and back sections to form a slit, thereby allowing the display to conform to the curved shape of the fingertip. With 128 electrodes per fingertip, the device represents the first implementation capable of covering the entire fingertips with stimulation points, thereby enabling tactile presentations on two fingertips.

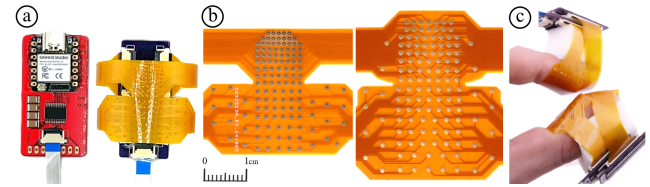


Figure 3: (a) Appearance of the device (left: main circuit board, right: flexible circuit board at the index finger); (b) Created Electrodes (left: Index finger, right: Thumb); (c) Example of an attached view.

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